Problem Set 4

QTM 220

Due 20 October at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

NBA Draft Data: Sample

The data set nba.sample.draft includes the following variables from the 2021-2022 NBA season:

- playername: name of player
- teamabbreviation : name of the team the player played on at the end of the season
- age: age of player
- playerheight : height in cm
- playerweight: weight in kg
- college: name of college player attended
- country : name of country where player was born
- draftyear : year player was drafted ('Undrafted' if undrafted)
- draftround : draft round player was selected
- draftnumber : number selected in his draft round ('Undrafted' if undrafted)
- gp: games played through season
- pts: average points scored
- reb: average rebounds
- ast: average assists
- netrating : team's point differential per 100 possessions while player is on the court

- orebpct : percentage of available offensive rebounds grabbed
- drebpct: percentage of available defensive rebounds grabbed
- usgpct: percentage of team plays used (FGA + Possession Ending FTA + TO)/POSS
- tspct: shooting efficiency (PTS/(2*(FGA + 0.44*FTA))
- astpct : percent of teammate field goals assisted
- draftlevel : draft round player was selected with undrafted players coded as 3

You are interested in the average net rating, the team's point differential per 100 possessions while the player is on the court, among different draft levels– players drafted in the first round, second round, or undrafted.

(a) What is the subsample mean net rating among those drafted in the first round? Second round? Undrafted? Create a scatterplot with draft level on the x-axis and net rating on the y-axis.

(b) Using the notation from class, show that the subsample mean net rating estimator among those drafted in the first round is unbiased for the subpopulation mean net rating among those drafted in the first round.

(c) What is the average incremental difference in mean net rating across the draft levels in your sample using the draftlevel variable,

$$\frac{1}{2}\sum_{x=2}^{3}\{\hat{\mu}(x) - \hat{\mu}(x-1)\}$$

(d) Rewrite part (c) as a linear combination of sub-sample means, $\sum_x \alpha(x)\hat{\mu}(x)$. What is your $\alpha(x)$ function?

(e) What is the estimated standard error of your average incremental difference using plug-in/normal approximation?

(f) Generate a bootstrap estimated sampling distribution of the average incremental difference estimator. Calculate a 95% confidence interval using your bootstrapped distribution.

(g) What is the average incremental difference in mean net rating across the draft levels in your sample weighted by the proportion of players drafted in each round (and undrafted)?

$$\sum_{x=2}^{3} P_x\{\hat{\mu}(x) - \hat{\mu}(x-1)\}$$

where $P_x = \frac{N_x}{\sum_{x=2}^{3} N_x}$

(h) Rewrite part (g) as a linear combination of sub-sample means, $\sum_x \hat{\alpha}(x)\hat{\mu}(x)$. What is your $\hat{\alpha}(x)$ function?

(i) What is the lower bound on the estimated standard error of your sub-sample weighted average incremental difference using plug-in/normal approximation?

(j) Generate a bootstrap estimated sampling distribution of the weighted average incremental difference estimator. Calculate a 95% confidence interval using your bootstrapped distribution.

(k) What is the least squares difference in mean net rating across draft levels in your sample?

$$\sum_x \tilde{w}(x)\hat{\mu}(x)$$

where $\tilde{w}(x) = \frac{N_x(x-\bar{X})}{\sum_i (X_i-\bar{X})^2}$

(l) Rewrite part (k) as a linear combination of sub-sample means using the $\alpha(x)$ notation, $\sum_x \hat{\alpha}(x)\hat{\mu}(x)$. What is your $\hat{\alpha}(x)$ function?

(m) What is the lower bound on the estimated standard error or your least squares summary of net rating across draft levels in your sample using plug-in/normal approximation?

(n) Generate a bootstrap estimated sampling distribution of the least squares estimator of net rating across draft levels. Calculate a 95% confidence interval using your bootstrapped distribution.

(o) Fill in the following table with information using your sub-sample estimates

	x	1	2	3
	N_x $\hat{\mu}(x)$ $\hat{\sigma}^2(x)$			
avg. increment	$\alpha(x)$ $\alpha^2(x)$ $\frac{\hat{\sigma}^2(x)\alpha^2(x)}{N_x}$			
weighted increment	$\hat{\alpha}(x)$ $\hat{\alpha}^2(x)$ $\frac{\hat{\sigma}^2(x)\hat{\alpha}^2(x)}{N_x}$			
least squares	$\hat{\alpha}(x)$ $\hat{\alpha}^2(x)$ $\frac{\hat{\sigma}^2(x)\hat{\alpha}^2(x)}{N_x}$			

Population Data

The data set nba_population includes the following variables from the 2021-2022 NBA season:

- playername: name of player
- teamabbreviation : name of the team the player played on at the end of the season
- age: age of player
- playerheight : height in cm
- playerweight: weight in kg

- college: name of college player attended
- country : name of country where player was born
- draftyear : year player was drafted ('Undrafted' if undrafted)
- draftround : draft round player was selected
- draftnumber : number selected in his draft round ('Undrafted' if undrafted)
- gp: games played through season
- pts: average points scored
- reb: average rebounds
- ast: average assists
- netrating : team's point differential per 100 possessions while player is on the court
- orebpct : percentage of available offensive rebounds grabbed
- drebpct: percentage of available defensive rebounds grabbed
- usgpct: percentage of team plays used (FGA + Possession Ending FTA + TO)/POSS
- tspct: shooting efficiency (PTS/(2*(FGA + 0.44*FTA))
- astpct : percent of teammate field goals assisted
- draftlevel : draft round player was selected with undrafted players coded as 3

(a) What are the sub-population mean net ratings among those drafted in the first round, second round, and undrafted? How do the sub-population means compare to your sub-sample means? How many players in the population are in each draft level? What is $\sigma^2(x)$ for each draft level? Fill in the table to organize your results.

x	1	2	3
m_x $\mu(x)$ $\sigma^2(x)$			

(b) What is the average incremental difference in mean net rating across the draft levels in the population using the draftlevel variable,

$$\frac{1}{2}\sum_{x=2}^{3}\{\mu(x)-\mu(x-1)\}$$

How does this compare to your estimate?

(c) Create a sampling distribution of your average incremental difference estimator. What size (n) should your repeated samples from the population be? What is the standard error for this estimator? Visually compare (with histograms) the sampling distribution to your bootstrap estimated sampling distribution for this estimator.

(d) What is the average incremental difference in mean net rating across the draft levels in the population weighted by the proportion of players drafted in each round (and undrafted)?

$$\sum_{x=2}^{3} p_x\{\mu(x)-\mu(x-1)\}$$

where $p_x = \frac{m_x}{\sum_{x=2}^{3} m_x}$ How does this compare to your estimate?

(e) Create a sampling distribution of your weighted average incremental difference estimator. What size (n) should your repeated samples from the population be? What is the standard error for this estimator? Visually compare (with histograms) the sampling distribution to your bootstrap estimated sampling distribution for this estimator.

(f) What is the least squares difference in mean net rating across draft levels in the population?

$$\sum_{x} \tilde{w}(x)\mu(x)$$

where $\tilde{w}(x) = \frac{m_x(x-\bar{X})}{\sum_i (X_i-\bar{X})^2}$ How does this compare to your estimate?

(g) Create a sampling distribution of your least squares estimator. What size (n) should your repeated samples from the population be? What is the standard error for this estimator? Visually compare (with histograms) the sampling distribution to your bootstrap estimated sampling distribution for this estimator.

(h) Overall, how did your estimates compare to the population? How similar were your estimated standard errors to the standard errors of your estimators? Why might the estimators have different standard errors?